

CLINICAL SLEEP EEG STUDY

ScoringNidra | AnalyseNidra quantitative report

Page 1 of 5

KHATUN_NARGIS_(1)_analyse_regional.csv

Study site: Neurophysiology Laboratory | Investigator: Study Investigator

Subject: Preview Subject | Research report preview

TRT 1195.5	TST 243.0	WASO 945.5
SOL 7.0	Sleep efficiency 20.3%	Maintenance efficiency 20.4%

Sleep architecture

Wake		952.5 min (79.7%)
N1		26.5 min (2.2%)
N2		192.0 min (16.1%)
N3		24.5 min (2.0%)
REM		0.0 min (0.0%)

Full-night hypnogram

Event	
W	
REM	
N1	
N2	
N3	

Stage latency dashboard (minutes from recording start)

Wake onset 0.0	N1 latency 7.0	N2 latency 9.0	Deep sleep latency 250.5	REM latency -
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Stage resilience and fragmentation

Stage	Longest	Mean streak	Median streak	Continuity profile
W	376.500	9.621	0.500	
N1	2.000	0.779	0.500	
N2	15.000	2.233	0.500	
N3	8.500	2.042	0.750	
REM	-	-	-	

Macrostructure values are reported in minutes unless otherwise specified.

Spindle and slow-wave morphometry with phase-amplitude coupling

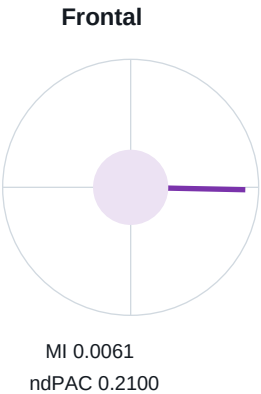
Regional spindle and slow-wave morphometry

Region	Spindles	Density/min	Spindle Hz	Slow waves	SW PTP μ V	SW slope
Frontal	44	0.2	13.2	173	109.2	205.5

Phasic spindle-slow wave coupling

Region	PAC MI	ndPAC	Spindle carrier Hz	SW driver Hz	Sigma peak phase (rad)
Frontal	0.0061	0.2100	14.8	1.0	-0.0

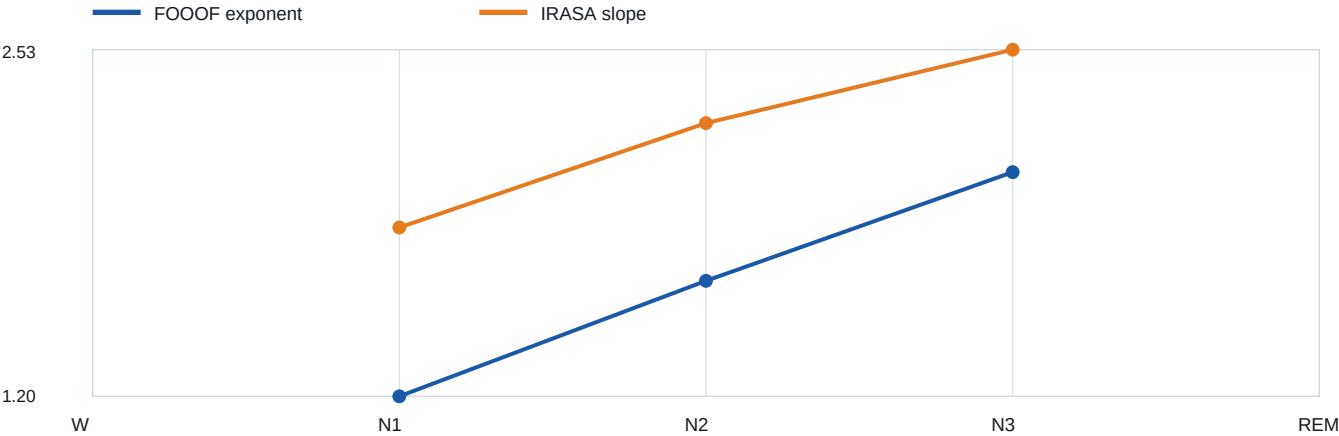
Spatial coupling comparison



Polar vectors indicate sigma peak phase on the slow-wave cycle. Rightward = 0 rad; upward = $\pi/2$ rad.

FOOOF and IRASA parameterization separates periodic peaks from the 1/f background

Stage-wise aperiodic trajectory (regional mean)



Aperiodic parameterization and model validation

Stage	FOOOF offset	FOOOF exponent	FOOOF error	IRASA intercept	IRASA slope	IRASA R2	IRASA AUC
W	-	-	-	-	-	-	-
N1	1.178	1.200	0.080	4.217	1.845	0.722	-62.330
N2	1.685	1.642	0.084	5.376	2.244	0.834	-109.922
N3	2.169	2.057	0.094	6.189	2.525	0.903	-204.633
REM	-	-	-	-	-	-	-

True oscillatory peaks after 1/f removal (regional mean)

Stage	Peak 1 CF	Peak 1 power	Peak 1 BW	Peak 2 CF	Peak 2 power	Peak 2 BW
W	-	-	-	-	-	-
N1	7.014	0.447	2.925	15.391	0.346	2.508
N2	6.589	0.481	3.023	12.547	0.374	2.921
N3	9.367	0.454	3.421	15.436	0.380	2.713
REM	-	-	-	-	-	-

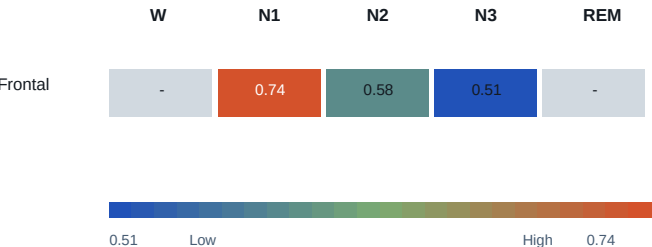
Interpretive quality flags

W Insufficient data	N1 Moderate confidence	N2 High confidence	N3 High confidence	REM Insufficient data
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Information theory, fractal dynamics, long-range correlations, compression, and temporal memory

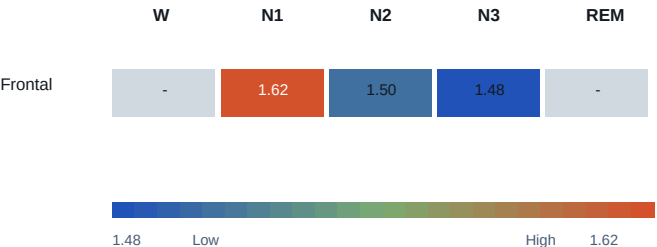
Signal entropy profile

Stage mean across metrics



Fractal dimension and LRTC

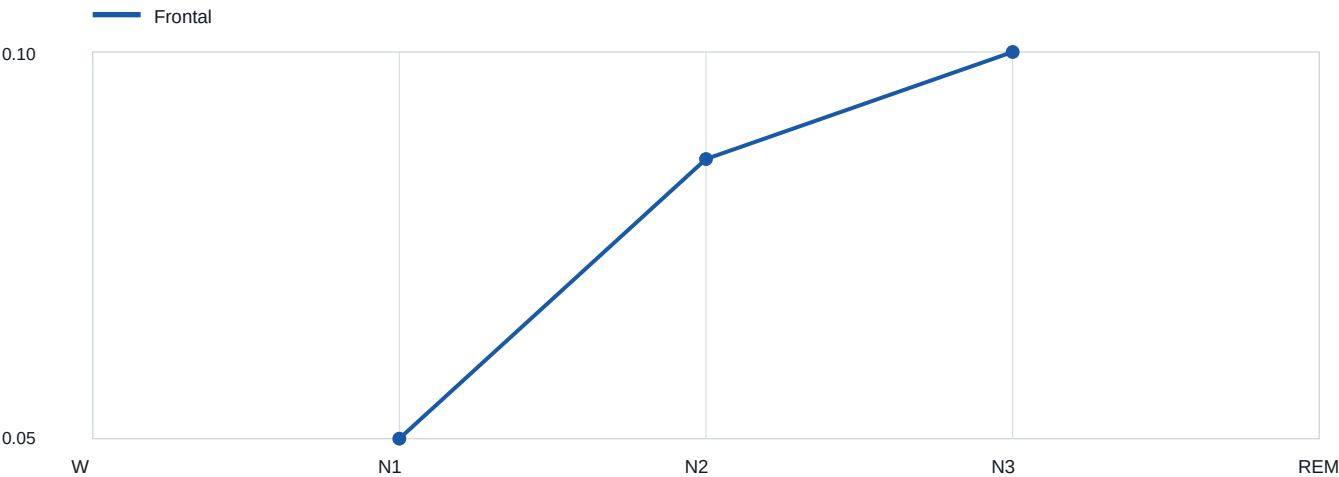
Stage mean across metrics



Information compression: global LZc versus stage-wise Lempel-Ziv

Region	Global LZc	W	N1	N2	N3	REM
Frontal	0.158	-	0.452	0.323	0.237	-

Autocorrelation window: temporal processing memory



Higher entropy and fractal metrics indicate less predictable dynamics. DFA summarizes long-range temporal correlations; ACW estimates temporal integration persistence.

A plain-language guide to the measurements in this report

Sleep amount and continuity

Estimated total sleep time was 243.0 minutes, with sleep efficiency of 20.3%. sleep-onset latency 7.0 min, wake after sleep onset 945.5 min, longest sleep-stage streak N2 15.0 min. A meaningful part of the recorded period was spent awake or transitioning between stages. Sleep efficiency alone does not identify the reason for wakefulness or fragmentation.

Sleep stages

N2 192.0 min (79.0%), N1 26.5 min (10.9%), N3 24.5 min (10.1%), REM 0.0 min (0.0%). N2 was the largest sleep-stage component in this recording. N3 onset 250.5 min. These are within-recording observations, not age-adjusted reference classifications.

Spindles, slow waves, and their timing

mean spindle density 0.20/min, slow-wave density 0.71/NREM min, PAC MI 0.0061, ndPAC 0.2100, sigma-peak phase -0.02 rad. The highest regional PAC MI was in Frontal (0.0061). Phase direction depends on the analysis convention; regional differences describe this recording and are not diagnostic thresholds.

Background spectrum and oscillatory peaks

FOOOF exponent was lowest in N1 (1.200) and highest in N3 (2.057). IRASA slope was lowest in N1 (1.845) and highest in N3 (2.525). IRASA R2 ranged from 0.722 to 0.903; lower-fit stages warrant more caution. The direction summarizes stage differences within this subject after regional averaging.

Complexity and temporal memory

permutation entropy was lowest in N3 (0.772) and highest in N1 (0.812). DFA was lowest in N1 (1.143) and highest in N3 (1.444). ACW was lowest in N1 (0.045) and highest in N3 (0.104). Global LZc was 0.158. Higher entropy indicates less predictable signal structure, while ACW reflects longer or shorter temporal persistence; neither direction is inherently better.

Important limitations

This automated report summarizes Frontal. Automated staging and qEEG measurements can be affected by artifacts, electrode quality, montage selection, and missing data. Results should be reviewed with the original recording by a qualified sleep or EEG professional and should not be used alone for diagnosis or treatment.